Kebei Jiang

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Profile

Applied scientist with 10+ years of experience in AI/ML research, quantitative experimentation, and end-to-end systems. Recent work spans LLM/VLM development, agent evaluation, synthetic benchmarks, LLM-as-a-judge calibration, RAG, multimodal document understanding, and model fine-tuning. Experienced in developing evaluation methodologies, designing controlled experiments, building production-oriented AI systems, and solving ambiguous problems across research and applied settings.

Core Expertise

AI/ML: LLMs · VLMs · Generative AI · Model fine-tuning · RAG · Agentic AI · Deep Learning · NLP
Evaluation & Data: LLM-as-a-judge calibration · Agent evaluation · Synthetic data · Evaluation benchmarks · Experiment design · Statistical modeling · Data analysis
Technical: Python · R · SQL/MySQL · PyTorch · Hugging Face · spaCy · vLLM · FastAPI · Docker · GitHub · Azure · GCP

Experience

Microsoft Jan 2020 – Sep 2026
Senior Applied Scientist

- Developed and deployed LLM and VLM solutions spanning agentic AI, RAG, knowledge mining, information extraction, prompt engineering, and model fine-tuning.
- Designed evaluation methodologies and experiments for LLM-based agents and AI systems, including synthetic benchmarks, behavioral diagnostics, quantitative metrics, and regression evaluation.
- Developed production ML/AI systems in Python, with experience in FastAPI, Docker, automated testing, production monitoring, and collaborative code reviews.
- Built end-to-end ML and information-processing systems from data preparation and feature engineering through model development, evaluation, and deployment.
- Led technical development across scientists, PMs, engineers, architects, and customer engineers, including technical design, implementation reviews, technical decisions, and customer engagements.

Viridien Jan 2015 – Jan 2020
Applied Scientist

- Applied statistical and machine-learning methods to process and model large-scale (TB) seismic data.
- Developed CNN-based solutions for seismic interpretation, including pixel-level identification of subsurface salt structures.

Selected Projects & Research

Work Projects

Specification-to-Calibration: Synthetic Supervision for LLM Judges *2026*

- Developed a method for automating LLM judge calibration: compiled natural-language specifications into coverage-controlled positive/negative examples, using targeted behavioral contrasts to revise rubrics while holding the judge model fixed.
- Improved position-debiased accuracy by 3.58 ± 0.58 pp on LLMBar; feedback-conditioned and GEPA-based calibration also improved balanced accuracy on BigCodeBench.

Period-Closing Skill — Stateful, Multi-Turn Agent Evaluation *2026*

- Designed 30+ reusable scenario templates covering diverse failure modes for an LLM agent that automates period-closing workflows involving due dates, calendars, worker assignment, and task dependencies; reviewed cases with the PM to validate representative scenarios.

- Parameterized scenarios across variables such as company names, task titles, worker names, dates, schedules, and dependencies to create diverse concrete test instances without manually designing every case.
- Built realistic multi-turn test environments with pre-existing state, calendar constraints, dependencies, and workers with similar names; fed simulated user responses back into the agent to test multi-turn behavior.
- Separated execution-trace quality from final task success and added seeding, trace, and task verification across repeated runs; evaluated TSR, tool-call accuracy, latency, token cost, and consistency.

Account Reconciliation Agent — Synthetic Benchmark & Adaptive Preference Monitoring *2026*

- Built an evaluation framework for an LLM agent resolving 400+ reconciliation exceptions between ledger and subledger using synthetic accounting workflows with ground-truth resolutions and tool-use paths.
- Identified an unrealistic uniform distribution of matching cases, validated the expected distribution with the PM and statistical analysis, and recalibrated synthetic data generation; used the resulting benchmark as a fixed offline regression set.
- Measured task success, tool-call accuracy/order/count, latency, and run-to-run consistency across repeated evaluations.
- Designed an adaptive preference-monitoring POC using synthetic data with shifting user-preference distributions: trained a tree-based model on a previous window of exception features and user-selected actions, evaluated it on the current window, and triggered retraining when accuracy fell below a PM-approved 90% threshold.

VLM Fine-Tuning — Taxonomy Generalization & Shortcut-Learning Diagnostic *2025*

- Designed a hierarchical taxonomy spanning hundreds of fashion-item classes and aligned it with GPT-4o's understanding of retail products; generated ∼1M synthetic training examples and fine-tuned an open-source VLM with LoRA, SFT, and DPO.
- Designed a controlled diagnostic to investigate what the model learned: after updating class names such as "sneaker" to "trainer," the fine-tuned model failed to adopt the changes while the base model adapted, exposing taxonomy memorization.
- Achieved ∼90% accuracy against GPT-4o ground truths; used the diagnostic to motivate separating domain adaptation from taxonomy alignment through distinct training strategies.

Multimodal RAG — Financial Insights Copilot *2024*

- Built an end-to-end RAG system for content-rich financial PDFs, spanning document parsing, multimodal indexing, embeddings, retrieval, query decomposition, and knowledge-graph exploration; released as part of Microsoft Financial Insights Copilot.
- Identified table-parsing issues in complicated financial documents and explored normalization strategies for downstream processing; systematically evaluated BM25 configurations and reranking models, improving offline retrieval by ∼20%.
- Explored separate text/table/image indices, ranking fusion, CLIP/SigLIP cross-modal matching, LlamaIndex query decomposition, and knowledge-graph/tree-based approaches including RAPTOR, KPG, and GraphRAG.
- Fine-tuned a Phi-3 Vision model on public chart datasets for visual question answering.

End-to-End Knowledge Mining — Multimodal Document Processing *2022–2023*

- Led development of an end-to-end knowledge-mining solution for large multimodal document collections in the energy domain using Azure Cognitive Search.
- Designed and optimized indexing flows and custom AI skills spanning chart detection, document classification, entity recognition, and summarization; led a team of scientists and collaborated with PMs, customer engineers, and customers.
- Identified unnecessary intermediate image storage as a major document-processing bottleneck and redesigned the flow to detect and index image bounding boxes before cropping directly from raw documents, reducing parsing time by ∼80%.
- Served AI skills for customer integration and supported technical questions and handoff.

Rule-Based NER with spaCy *2021*

- Developed a rule-based entity extraction system for geo-location entities in Oil & Gas documents, including class identification and post-processing to reduce false positives from overlapping and ambiguous names.
- Achieved over 80% accuracy and implemented the solution as a custom skill in Azure Cognitive Search.

Automatic Seismic Salt Interpretation *2018*

- Used a U-Net architecture to identify salt structures at pixel level in seismic data.
- Implemented stratified folding, ELU, and Lovasz Loss to improve precision; work was accepted by the 3rd International Conference on Information System & Data Mining.

Personal Projects

OptiBuild — Domain-Agnostic Optimization Agent | Personal Project *2026*

- Built a domain-agnostic multi-agent system that converts natural-language optimization requests into structured Operations Research models, evaluates model completeness and intent fidelity, and solves them using CP-SAT with TOPSIS-based multi-objective ranking.
- Designed a staged modelization and repair loop with Pydantic contracts, deterministic validation, LLM-based intent evaluation, structured execution traces, and failure handling for missing data, infeasible models, and clarification needs.
- Implemented automated testing and security controls, including 127 offline tests, PII redaction, an explicit safety gate, and constrained handling of LLM-generated formulas, queries, and data-cleaning operations.

Selected Technical Learning

CS336: Language Modeling from Scratch, Stanford *2025*

- Implemented major components of language-model development, including tokenization, FlashAttention2 in Triton, data filtering, and pre-/post-training methods including GRPO.

Neural Networks: Zero to Hero, Andrej Karpathy *2023*

- Implemented a Transformers-based English–Spanish translator from scratch and trained it on the Helsinki-NLP/opus-100 dataset.

Reinforcement Learning Specialization, Coursera *2023*

- Completed the specialization to deepen understanding of reinforcement learning motivated by the use of RLHF in LLM development.

Education

Louisiana State University, Baton Rouge, LA 2007 – 2014
PhD, Physics

Publications

- Guang Yang, Kebei Jiang, David P. Lundquist, Prerna Singh. *Specification-to-Calibration: Synthetic Supervision for LLM Judges.* Submitted to JUDGe @ NeurIPS 2026.
- Moschoula Pternea, Prerna Singh, Abir Chakraborty, Yagna Oruganti, Mirco Milletari, Sayli Bapat, Kebei Jiang. *The RL/LLM Taxonomy Tree: Reviewing Synergies Between Reinforcement Learning and Large Language Models.* Journal of Artificial Intelligence Research, 2024. DOI: 10.1613/jair.1.15960.
- Yu Zeng, Kebei Jiang, Jie Chen. *Automatic Seismic Salt Interpretation with Deep Convolutional Neural Networks.* Proceedings of the 2019 3rd International Conference on Information System & Data Mining, pp. 16–20.

Patent

- Yinheng Li, Kebei Jiang. *Concurrent Labeling of Sequences of Words and Individual Words.* U.S. patent application, MS# 411939-US-NP, 2022.